MILITARY SPECIFICATION

LIFE PRESERVER SUPPORT PACKAGE FOR

LIFE PRESERVER, MK 4

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for construction and performance for MK 4 life preserver, repair kit and spare parts and test kit for the Life Preserver Support Package.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-285 - Thread, Polyester.
QQ-A-591 - Aluminum Alloy Die Castings.
QQ-B-626 - Brass, Leaded and Nonleaded: Rods, Shapes, Forging Flat Products with Finished Edges (Bar and Strip).
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought.
QQ-W-321 - Wire, Copper Alloy.

MILITARY

MIL-L-3454 - Life Preservers: Packaging of.
MIL-R-6855 - Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes.
MIL-I-6869 - Impregnants for Aluminum Alloy and Magnesium Alloy Castings.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3242, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY - (Continued)

MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-C-16385 - Cartridge, Carbon Dioxide, Inflatable Life Preserver Use.
MIL-M-19595 - Magnetic Effect Limits for Nonmagnetic Equipment Used in the Proximity of Magnetic Influence Ordnance.
MIL-C-25369 - Cylinder, Carbon Dioxide, for Inflation of Pneumatic Life Preservers.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

2.1.2 Other Government drawings and publications. The following Government drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND

592-5403167 - Life Preserver, Mark 4 Main Assembly.
592-5403168 - Life Preserver, Mark 4 Outer Cover Assembly.
592-5403169 - Life Preserver, Mark 4 Bladder Assembly, Inner.
592-5403173 - Life Preserver, Mark 4 Stem Assy, Inflator.
592-5403179 - Kit, Repair.
592-5403181 - Life Preserver Support Package - Life Preserver, Mark 4 (Assembly).
DL 592-5403181 - Life Preserver Support Package - Live Preserver, Mark 4 (Data List).
592-5412454 - Kit Spare Parts/Test-Life Preserver, Support Package.
803-1411524 - CO_2 Holder and Holder Gage Details.
803-2274038 - Life Preserver Yoke Type SCUBA USN MK3 Duel Inflator Assembly and Details.

PUBLICATIONS

UNITED STATES COAST GUARD

Code of Federal Regulations, Title 46, Sub-Chapter II
160.060 - Marine Buoyant Devices.
160.064 - Marine Buoyant Devices.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions

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should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
B 117 - Standard Method of Salt Spray (Fog) Testing. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, (see 6.2), the contractor shall furnish one complete Life Preserver Support Package for first article inspection and approval (see 4.3 and 6.2.1).

3.2 Material. The life preserver shall be constructed of materials as specified on the applicable drawings and as specified herein. In case of conflict between the requirements specified herein and the requirements specified on the referenced drawings, the requirements of this specification shall govern.

3.2.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.2 Nonmagnetic material. The life preserver shall be constructed of material which is capable of withstanding the magnetic effects test of 4.9.13.

3.2.3 Mechanical inflation system.

3.2.3.1 Inflator assemblies. Each life preserver shall be provided with an assembly consisting of two dual inflators conforming to drawing 803-2274038. The assembly shall consist of holders for use with CO_2 cartridges conforming to type II of MIL-C-16385 or type I of MIL-C-25369. The

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holder shall be a high pressure aluminum die casting conforming to alloy no. A380 in accordance with QQ-A-591. The body shall be anodized in accordance with type II, class 1 of MIL-A-8625 and then impregnated in accordance with class 1 of MIL-I-6869. The cartridge shall be punctured and the piercing pin shall be retracted with a travel of the actuating lever through an arc of not over 180 degrees. The discharge mechanism shall operate on the eccentric principle which will allow the piercing pin to return to the original position after operation, even when the lever is held in the tripped position. The piercing pin shall be sealed so that there shall be no leakage of gas from the holder when the discharge lever is in any position. The piercing pin in the normal position shall be 0.010 inch behind the cartridge seat to prevent accidental puncture. Clearance shall be provided between the open end of the holder and the CO_2 cartridge, even in the event that the seating surface, where the puncturable end of the cartridge seats within the holder becomes worn, so that the cartridge can be screwed into the holder to a greater depth and onto the seat. Holders shall withstand a hydrostatic pressure of 3,500 pound per square inch gage (lb/in^2g).

3.2.3.2 Preserver holder assemblies. The preserver holder assemblies shall have levers and setscrews fabricated from unplated brass conforming to QQ-B-626. There shall be provisions for securing the actuating levers against the sides of the holders by a length of 0.016-inch diameter safety wire. The holder threads shall be cold formed, rolled thread, and shall be coated with a silicone grease after the holders have been assembled on the life preserver. There shall be hard drawn phosphor-bronze springs made of wire conforming to QQ-W-321 to provide that the piercing pins separate from the cartridges immediately after puncturing to preclude any possibility of delay in inflation by the pins remaining in and sealing the holes. The piercing pins shall be hardened (nickel-copper-aluminum alloy) conforming to class A of QQ-N-286.

3.2.3.3 Inflation lanyard. The inflation lanyard assembly shall be as detailed on drawing 592-5403167. When tested in accordance with 4.9.15, the lanyard assembly shall not break or fail at less than a minimum breaking strength of 50 pounds.

3.2.3.4 Inflator stem. The holder shall be securely attached to the life preserver by means of an uncoated brass inflator stem conforming to drawing 592-5403173. Two synthetic rubber gaskets conforming to class 2, type A, grade 80 of MIL-R-6855 shall be provided, one for each side of the holder attachment. The inflator stem shall provide a zero leakage connection between the inflation assembly and the life preserver.

3.2.4 Color. Life preserver parts and materials shall have exterior color in accordance with number 37038 of FED-STD-595.

3.2.5 Thread. Thread used for all sewing shall conform to type II, class 1, size F of V-T-285 and shall be colored to match the items sewn.

3.2.6 Bladder. The material used for the bladder shall be a polyether base, polyurethane film capable of being dielectrically heat sealed to itself. The film shall be fungus resistant and contain no waxes or other ingredient which adversely affect its dielectric sealability. The film shall be of a clear color with a matte finish on both sides. The

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material shall have air-holding characteristics and be free of pin holes, surface imperfections, and particles of foreign matter. The film shall meet the physical requirements of table I. A certificate of compliance from the material supplier shall be satisfactory for meeting these requirements.

TABLE I. Physical requirements of polyurethane film.

Characteristic	Requirement
Specific gravity	1.1
Hardness	80 Shore A (Typical)
Brittle temperature (masland SPI cold crack)	-70°F max.
Tensile strength (ultimate)	3000 lb/in^2 min.
Ultimate elongation (at break)	500 percent min.
Tear strength (graves)	400 lb/in^2 min
Abrasion resistance: (taber) 1000 Cycles, CS17 Wheel; Load 1000 gram Percent weight loss	0.06 percent max.

3.2.7 Outer covering. The outer covering shall conform to the requirements on drawing number 592-5403168 and meet the physical requirements of table II. A certificate of compliance shall be provided in accordance with the data ordering document included in the contract (see 6.2.2).

TABLE II. Physical requirements of outer covering.

Characteristic	Requirement
Burst strength (Mullen Hydrostatic)	600 lbs min
Tensile strength (grab method)	500 warp x 400 fill lbs min
Elongation at break	37 x 32 percent max
Tear (tongue method)	20 x 20 lbs min
Puncture resistance	70 lbs min
Tear (trapezoid method)	30 x 20 lbs min
Blocking scale rating	No. 2

3.2.8 Webbing. The webbing shall meet the performance requirements for webbing of Coast Guard Specifications 160.060 and 160.064. A certificate of compliance shall be provided in accordance with the data ordering document included in the contract (see 6.2.2).

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3.2.9 Corrosion resistance. The oral valve, relief valve, CO_2 inflation devices and all metal parts shall be made of corrosion-resistant material and shall show no evidence of corrosion after subjection to the 100 hour salt spray test of 4.9.12.

3.3 General requirements.

3.3.1 Design, manufacture and assembly. The manufacturing and assembly details of the life preservers shall conform to the drawings listed on DL 592-5403181.

3.3.2 Heat sealing. In the attachment of all parts and the construction of the seams of the inner bladder, a radio frequency dielectric heat sealing process shall be used. The seams of the inner bladder shall be as shown on drawing 592-5403169. The seams shall be capable of withstanding a 20 pounds per inch (lb/in) peel load when tested in accordance with 4.9.11.

3.3.3 Marking. Identification marking shall be by silkscreen method using white ink. The marking shall be waterproof and cause no deterioration of fabric.

3.3.3.1 Serial numbers. Serial numbers will be furnished by the contracting activity. Serial numbers shall be located in accordance with drawing number 592-5403168.

3.3.4 Repair kit. Each life preserver shall be provided with a repair kit consisting of the items listed on drawing 592-5403179.

3.3.5 Spare parts/test kit. Two spare parts/test kits shall be provided. Each spare parts/test kit shall consist of the items listed on drawing 592-5412454.

3.4 Reliability. The life preserver shall be constructed in a manner that will ensure a 97 percent operational reliability with a 95 percent confidence level under all practical combinations of storage and operating environments (see 4.9.14).

3.5 Operation. When and inflation mechanism containing two CO_2 cartridge is released into the life preserver the preserver shall fully inflate in not more than 5 seconds.

3.6 Hydrostatic pressure. The life preserver shall operate properly after being subjected to the hydrostatic pressure cycle test specified in 4.9.1.1.

3.7 Impact. The life preserver shall be capable of withstanding an impact of 18 feet per second (ft/sec).

3.8 Buoyancy. The life preserver shall be capable of supporting a lead weight, weighing 42 pounds in air, without sinking.

3.9 Operation of oral valve.

3.9.1 During assembly. The purge ports of the oral valve shall operate properly before and after the button is depressed.

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3.9.1.2 After assembly. After inflation through the oral valve the purge ports shall prevent water from entering the inner bladder. When the preserver is inflated until the relief valve opens, the oral valve shall operate without difficulty and shall not leak when the preserver is fully inflated. The oral valve shall work without difficulty and shut off securely when the preserver is deflated.

3.10 Pressure. The preserver shall be capable of withstanding a pressure test of 10 lb/in^2g after the over-pressure relief valve is removed and the opening capped.

3.11 Leakage.

3.11.1 Twenty-four hours. The life preserver shall have an internal pressure corrected for barometric pressure and temperature of not less than 1.0 lb/in^2g after being inflated to 1.5 lb/in^2g and allowed to stand for 24 hours.

3.11.2 Six hours. The life preserver shall remain firm after being inflated until the relief valve opens and allowing to stand for 6 hours.

3.12 Operation of relief valve. The relief valve shall relieve at a pressure of 2.5 $\pm$ 0.5 lb/in^2g when tested in accordance with 4.9.6.

3.13 Oral inflation hose.

3.13.1 Porosity. The oral inflation hose shall show no evidence of leakage due to porosity nor weakening of the hose as revealed by excessive deformation in any portion of the hose when subjected to an internal air pressure of 15 lb/in^2g for a minimum of 10 seconds while submerged under water.

3.13.2 Tensile strength. The tensile strength of the oral inflation hose shall not be less than 50 pounds when tested in accordance with 4.9.9.2.

3.14 Holding strength (oral inflation assembly). The load required to separate the hose from the oral inflation valve or oral inflation elbow shall be not less than 50 pounds (see 4.9.10).

3.15 Workmanship. Workmanship shall be such as to ensure compliance with the requirements of this specification. Because of the emergency and life support use of the equipment, it is essential that the life preserver functions properly under conditions to which it may be subjected.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection system. The contractor shall develop and maintain an inspection system program plan acceptable to the Government for supplies and services covered by this specification in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.2 Classification of inspection. The examination and testing of the life preservers shall be classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be performed on the sample units as specified in 4.3.1. First article inspection shall consist of the examination and tests of table III. Failure of any examination or test specified shall be cause for refusal to grant first article approval.

TABLE III. First article inspection.[1/]

Inspections	Inspection paragraph	Number of sample units inspected
Visual examination	4.6	8
Operation	4.9.1	8
Buoyancy	4.9.2	8
Operation of oral valve	4.9.3.2	8
Pressure	4.9.4	8
Leakage	4.9.5.1	8
Operation of relief valve	4.9.6	8
Dimensional inspection (gage) of CO_2 inflation assembly	4.9.7	22
Leakage and operation (CO_2 inflation assembly)	4.9.8	22
Oral inflation assembly holding strength	4.9.10	4
RF heat sealing	4.9.11	1
Salt spray	4.9.12	1
Magnetic effect	4.9.13	3
Reliability	4.9.14	2
Breaking strength (lanyard assembly)	4.9.15	4
Inspection of packaging	4.10	8

[1/] Life preservers shall have been subjected to and passed the magnetic effect inspection of 4.9.13 before submission for first article inspection.

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4.3.1 Samples for first article inspection. Prior to starting regular production, the following samples shall be submitted for first article inspection:

(a) One complete life preserver support package, including properly charged CO_2 cartridges. (See drawing 592-5403181).
(b) Four pieces mesh material 2-3/4 inches wide x 8 inches long.
(c) Two yards each of all outer covering and urethane bladder material used in life preserver and preserver cover. The name of the supplier, (if different from the manufacturer) and part numbers shall be furnished with the samples.
(d) One complete inner bladder assembly.
(e) Two yards each of all fastening tape used on preserver.
(f) One cone (500 yards) of polyester sewing thread of the color used in the preserver.
(g) Four sets of all hardware, including CO_2 inflator system, and zippers used in life preserver. The name of the supplier (if different from the manufacturer), and part number shall be furnished with the samples.
(h) Four pressure relief valve assemblies and oral inflation assemblies including internal and external threaded access fittings and seal rings. The name of the supplier (if different from the manufacturer) and part number shall be furnished with the samples.

4.3.2 First article sample identification. Samples for first article inspection shall be forwarded to the contracting activity plainly identified by securely attached durable tags marked with the following information:

(a) Samples for first article inspection.
(b) Name of product, together with complete identifying data such as manufacturer's designation number.
(c) Name of manufacturer.
(d) Contract or order number.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the visual examination of 4.6, and the tests specified for groups A and B (see 4.4.1 and 4.4.2).

4.4.1 Group A tests. Group A tests shall be conducted in accordance with table IV as a part of the inspection system, (see 4.1.1) during the assembly of the life preserver. When specified by the contracting activity (see 6.2.1), tests shall be witnessed by the Government Inspector. Preservers not conforming to the tests shall be reworked prior to submission to any Group B inspection but under no circumstances shall an air retaining seam, manifold attachment or seal ring attachment be reworked (see 4.4.3). Failures of any samples of oral inflation hose or oral inflation assemblies shall be cause to rework these components. In adition, the contractor shall prepare RF heat sealed test samples from a test bladder and shall subject the samples to the test of 4.9.11. One bladder shall be so tested at the start of each day's production and every 50th bladder thereafter.

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TABLE IV. Group A tests.

Test	Quantity	Paragraph
Operation of oral valves	Each	4.9.3.1
Leakage - Six hour test	Each	4.9.5.2
Dimensional examination of inflation assembly	Each	4.9.7
Leakage and operation (inflation assembly)	Each	4.9.8
Oral inflation hose (porosity)	Six samples per 1000	4.9.9.1
Oral inflation hose (tensile)	Six samples per 1000	4.9.9.2
Oral inflation assembly (holding strength)	Six samples per 1000	4.9.10
RF heat sealing	1 every 50	4.9.11

4.4.2 Group B tests. Group B tests shall be conducted in accordance with table V on life preserver samples selected in accordance with 4.5.3 which have passed the visual examination tests (see 4.6) and the component tests of 4.4.1. When specified by the contracting activity (see 6.2.1), tests shall be witnessed by the Government Inspector. Any item in the sample containing one or more defects shall not be offered for delivery. If the number of nonconforming items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE V. Group B tests.

Test	Paragraph
Operation (CO_2 inflation)[1/]	4.9.1
Buoyancy[1/]	4.9.2
Operation of oral valve	4.9.3.2
Pressure	4.9.4
Leakage - 24 hour test	4.9.5.1
Operation of relief valve	4.9.6

[1/] The buoyancy test specified in 4.9.2 shall be conducted when the CO_2 cartridges are expended during the operation test of 4.9.1.

4.4.3 Rejected lots. Lots rejected under 4.4.2 may be reworked and resubmitted for examination as specified herein, except that life preservers shall be tested 100 percent for the defect or defects which caused the original rejection. Rejected lots may only be resubmitted for acceptance after complete screening for, and correction of noted defects.

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4.5 Sampling procedures.

4.5.1 Lot. A lot shall consist of all preservers submitted for inspection at one time. Maximum lot size shall not exceed 250 life preservers.

4.5.2 Sampling for visual examination (major and minor defects). A random sample of life preservers shall be slected from each lot in accordance with inspection level I of MIL-STD-105 for the visual examination for major and minor defects in 4.6.2. The acceptable quality level (AQL) shall be 4.0 for major defects and 10.0 for minor defects.

4.5.3 Sampling for group B tests. A random sample of life preservers shall be selected from each lot in accordance with the inspection levels shown in table VI as specified in MIL-STD-105 for tests specified in 4.6.2. The acceptance number shall be zero for all tests. Rejected preservers shall not be resubmitted.

TABLE VI. Inspection levels for group B tests.

Tests	Inspection level
Operation (CO_2 inflation)	I
Buoyancy	S-I
Operation of oral valve	I
Pressure	I
Leakage 24 hour test	I
Operation of relief valve	I

4.6 Visual examination. Life preservers shall be thoroughly visually examined to determine conformance with this specification and the applicable drawings.

4.6.1 For critical defects. The life preservers shall be examined 100 percent for critical defects as defined in table VII. Life preservers containing one or more critical defects shall not be offered for delivery.

4.6.2 For major and minor defects. Sample preservers selected in accordance with 4.5.2 shall be examined for major and minor defects (as defined in table VII). Any sample containing one or more defects shall not be offered for delivery, and if the number of nonconforming items exceeds the acceptance number for that sample, this shall be cause for rejection of the entire lot.

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TABLE VII. Classification of defects.

Categories	Defects
Critical:	
1	Buoyancy chamber material defective; evidence of splits, tears, weak spots, patched holes or fraying.
2	Buoyancy chamber seams defective, incomplete, or not as specified; evidence of bubbles, incomplete sealing, or seams in buoyancy chamber not the specified type; width of seams less than the required minimum.
3	Oral inflation valve defective or not secure in hose; hose evidences cracks, splits, bubbles, bulges, pin holes or other deleterious effects.
4	Oral inflation elbow or oral inflation hose not secure.
5	Pressure relief valves missing, defective, not as required, not properly installed and located.
6	Internal and external threaded access fittings defective, not properly installed, or improperly sealed.
7	Holder and inflation mechanism assembly defective, not securely attached, or not as specified, lanyard not securely attached to actuating lever, lever holding pin not staked in place.
8	Failure of specified gage to seat firmly against seating surface of holder, O-rings (for holder and piercing pin) missing, defective or not as required (examine prior to assembly).
9	Gaskets (each side of holder attachment) missing, defective, or not as required.
10	Zipper missing, defective, not as required, not properly installed and located.
11	Evidence of reworking air retaining seam or attachments of manifold or seal ring.
Major:	
101	Velcro for securing inflation hose defective or not securely attached as specified.
102	Velcro for covering zipper tab defective or not securely attached as specified.
103	Holder not secure; inflater cover not secure.
104	Harness webbing evidences tears, splits, fraying or weak spots.
105	Harness stitching defective, incomplete, or not as specified.
106	Harness hardware missing, defective, or not secure; evidence of cracks, deep pits, thin spots or sharp surfaces.
107	Holder actuating lever not safety wired as specified.
108	Metal parts (holder, lever or manifold) not finished as specified; finish defective or incomplete.
109	Date of manufacture marking missing, not permanent, not located as specified, illegible or incorrect.

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TABLE VII. Classification of defects. - Continued

Categories	Defects
Major: - Continued	
110	Set screw provided in holder, for handling CO_2, cartridge missing or not staked.
111	Holder threads not greased as required after assembly on life preserver.
Minor:	
201	Length of lanyard not as required.
202	Color of life preserver parts and materials not as specified.
203	Marking, manufacturer's name or trademark and identification missing, incorrect, illegible, not permanent or not located as specified.
204	Packaging and packing not as specified.
205	Stitching on outer cover not as specified.

4.6.2.1 Rejected lots. Lots rejected under 4.6.2 may be resubmitted for examination as specified in 4.6.2. Where the original acceptance number was zero, a sample size represented by the next higher sample size code letter shall be slected with the acceptance number remaining zero. The resubmitted lot shall be examined by the contractor and witnessed by the Government Inspector when specified (see 6.2.1) using tightened examination. Rejected lots may only be resubmitted for acceptance after complete screening and correction for noted defects.

4.7 Test conditions.

4.7.1 Atmospheric conditions. Unless otherwise specified herein all tests required by this specification shall be conducted at prevailing atmospheric pressure, a temperature of 77° + 18° Fahrenheit (°F), and a relative humidity of 95 percent, or less. When the inspections are conducted at an atmospheric pressure or temperature different from the values specified herein, proper corrections shall be made for the change in instrument readings. The instrument used shall contain no mercury.

4.7.2 CO_2 cartridge for test purposes. CO_2 cartridges used in conducting the tests specified in 4.9.1 and shall be in accordance with type I of MIL-C-25369.

4.8 Possible test failures. Possible test failures shall be determined as follows:

(a) Operation of assembled life preserver:
(1) Failure of life preserver to inflate when lanyard is pulled; evidence of leaks through bladder, seams, inflation assembly or piercing pin, O-rings, access

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fittings, over pressure valve or oral inflation hose and assembly.

(2) Life preserver cannot be readily inflated through the oral inflation valve, or there is evidence of leakage back through valve after inflation.

(3) Relief valves fail to relieve within the specified limits when the preserver is inflated as required.

(4) Evidence of material failure or parting of seams when the preserver is subjected to the specified pressure test.

(b) Leakage test (24 hour test): The pressure within the life preserver less than the minimum permissible after it has been inflated and then allowed to stand for the required length of time.

(c) Buoyancy: Failure of the inflated life preserver to support, without sinking, the specified weight.

(d) Breaking strength test: Discharge lever, rope, and button fail to withstand the required minimum pull.

4.9 Test methods. Where not specifically controlled by requirements herein, the contracting activity shall have approval authority for all test equipment and methods used in performance of tests and inspections required herein, in order to assure adequacy of product testing.

4.9.1 Operation (CO_2). The operation of the preserver shall be tested by releasing the CO_2 cartridges into the life preserver in the normal manner. The life preserver shall fully inflate in not more than 5 seconds.

4.9.1.1 Hydrostatic pressure. The uninflated life preserver shall be tested for normal operation immediately after passing through the following hydrostatic pressure cycle:

(a) With the life preserver immersed in water, increase hydrostatic pressure at a rate equivalent to a descent of 75 feet per minute (ft/min) until a seawater depth of 200 $\pm$ 10 feet is reached.

(b) Hold for 1 hour, then decrease pressure at a rate equivalent to an ascent of 60 ft/min to the surface.

4.9.1.2 Impact. The uninflated life preserver shall be tested for normal operation immediately after being dropped into seawater at a velocity of at least 18 ft/sec at impact.

4.9.2 Buoyancy. Immediately upon completion of the operation CO_2 test (see 4.9.1), the preserver shall be attached to a lead weight, the latter weighing 42 lbs in air. The preserver and weight shall then be placed in clear fresh water. The life preserver shall support this weight without sinking.

4.9.3 Operation of oral valve. The tests specified in 4.9.3.1 and 4.9.3.2 shall be conducted as specified in 4.3 and 4.4.

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4.9.3.1 During assembly. Pass air through the oral valve from the mouthpiece. Correct operation of the oral valve shall be verified such that the purge ports shall operate properly without the button depressed and the air shall pass through the oral valve properly after the button is depressed.

4.9.3.2 After assembly. The life preserver shall be inflated through the oral-valve. The purge ports shall operate properly to prevent water from entering the inner bladder. The preserver shall be inflated until the relief valve opens. The oral valve shall operate without difficulty and after the vest is fully inflated shall not leak. The life preserver shall then be deflated through the oral valve and it shall be ascertained that the valve works without difficulty and shuts off securely.

4.9.4 Pressure. With the over pressure valve removed and the test gage installed on the over pressure valve external threaded access fitting, the life preserver shall be inflated with air through the fitting on the test gage to a pressure of 10 lb/in^2 and shall be allowed to stand for a minimum of 10 minutes. The preserver shall then be charged and readjusted to 5 lb/in^2 and if the pressure at the end of a further 30 minute period, minimum is less than 4.5 lb/in^2 the preserver shall be rejected.

4.9.5 Leakage. The following leakage tests shall be performed on the completed life preserver as specified.

4.9.5.1 Twenty-four hour test. With the over pressure valve removed and the test gage installed on the over pressure valve external threaded access fitting, the life preserver shall be inflated with air through the oral inflation valve to a pressure of 1.5 $lb/in^2g \pm 0.2$ lb/in^2g and allowed to stand for a minimum of 15 minutes. The pressure shall then be checked and readjusted if necessary to the original pressure of 1.5 $lb/in^2g \pm 0.2$ lb/in^2g. The temperature and barometric pressure shall be recorded at this time. At the end of a minimum of 24 hours after the readjustment period, the pressure shall be measured and corrected for any changes in temperature or barometric pressure (see 4.8.1). The corrected pressure shall be not less than 1.0 lb/in^2g.

4.9.5.2 Six hour test. The completed life preserver shall be inflated until the relief valve opens and allowed to stand for a minimum of 6 hours. At the end of this required period the life preserver shall be inspected for firmness. Life preservers not firm shall be inspected for leaks, reworked and tested again.

4.9.6 Operation of relief valve. The oral inflation elbow shall be removed and the test gage installed on the oral inflation assembly threaded access fitting. The preserver shall then be inflated with air through the fitting on the test gage to test the operation of the relief valve. Each valve shall relieve at a pressure of 2.5 $lb/in^2g \pm 0.5$ lb/in^2g.

4.9.7 Dimensional inspection (gage) of inflation assembly. Holder assembly (complete with O-ring) shall be dimensionally checked with a plug gage conforming to the dimensions shown on drawing 803-1411524. Failure of the plug gage to seat firmly against seating surface (where the puncturable end of the cartridge seats within the holder) shall be cause for rejection.

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4.9.8 Leakage and operation (inflation assembly). Holders shall be tested for leakage under water with air at a pressure of 40 lb/in^{2}g, with the actuating lever in the actuated position. (The actuated position is considered to be the position of the lever after the piercing pin has punctured the cartridge and has completely retracted. Cartridges are not required to determine the actuated position.) Holders shall be subjected to a hydrostatic pressure of 3,500 lb/in^{2}g. Holders shall also be tested for leakage under water with the actuating lever slowly carried through one complete cycle.

4.9.9 Oral inflation hose. Before the oral inflation valve is installed in the oral inflation hose, the tests specified in 4.9.9.1 and 4.9.9.2 shall be conducted.

4.9.9.1 Porosity. The oral inflation hose shall be subjected to an internal air pressure of 15 lb/in^{2}g for a minimum of 10 seconds while submerged underwater. There shall be no evidence of air leakage due to porosity. There shall be no evidence of weakening of the hose as revealed by excessive deformation in any portion of the hose.

4.9.9.2 Tensile. The ends of the oral inflation hose shall be placed in the jaws of a test machine. The rate of load application shall be 10 in/min $\pm$ 2.5 in/min. The tensile strength shall be not less than 50 pounds.

4.9.10 Oral inflation assembly holding strength. This test shall be conducted after the oral inflation valve and oral inflation elbow have been attached and secured to the oral inflation hose. The elbow and valve shall be placed in the jaws of a test machine. The rate of load application shall be 10 in/min $\pm$ 2.5 in/min. The load to separate the hose from the oral inflation valve or oral inflation elbow shall not be less than 50 pounds.

4.9.11 RF dielectric heat sealing. Fifteen inspection samples 1 inch wide by 2 inches long shall be prepared from the test bladder as shown on drawing 592-5403169. Each inspection sample shall be placed in the jaws of a test machine so that the applied load will peel the seam. The rate of load application shall be 7.5 in/min $\pm$ 2.5 in/min. The peel strength of each inspection sample shall be not less than 20 lb/in.

4.9.12 Salt spray. A 100 hour salt spray test shall be conducted on a completed life preserver in accordance with ASTM B 117. The life preserver shall be tested for normal operation at the conclusion of the test.

4.9.13 Magnetic effect test. Prior to first article inspection, the following magnetic effect tests shall be conducted on hardware and complete life preservers:

(a) All hardware to be used in the fabrication of completed life preservers shall be submitted to the Navy Explosive Ordnance Disposal Technology Center (NEODTC), Attn: Code 45, Indian Head, MD 20640, for test in accordance with MIL-M-19595, prior to assembly into the preserver.

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(b) After notification of approval of hardware (item (a)), three complete life preservers for first article inspection, shall be submitted to NEODTC for test in accordance with MIL-M-19595. These samples may be included in the quantity of samples required for first article inspection (see 4.3.1) if they conform to the magnetic effect test.
(c) When only replacement hardware is acquired for life preservers, the hardware shall be submitted for test in accordance with 4.9.13(a).

4.9.14 Reliability. The life preserver shall be inflated until relief valve opens for 100 cycles at the surface. No failure shall occur.

4.9.15 Breaking strength (lanyard assembly). The breaking strength of the lanyard assembly shall be determined by securing the lever to the upper jaw of a test machine and the pull knob in the bottom jaw. The portion of the lever containing the cord shall be between the upper and lower jaws. The jaws shall be separated until failure occurs. The rate of load application shall be 10 in/min $\pm$ 2.5 in/min.

4.10 Inspection of packaging. The packaging, packing, and marking of life preservers shall be inspected to determine conformance with Section 5 herein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the requirements of referenced documents listed in section 2, see 6.5.)

5.1 Packaging, packing and marking. Life Preserver Support Packages complete with charged CO_2 cartridges shall be packaged level A or C, packed level A, B, or C as specified (see 6.2.1) and marked in accordance with MIL-P-3454.

6. NOTES

6.1 Intended use. The life preserver covered by this specification is intended for use as an emergency equipment by SCUBA personnel while operating under water at depths to 200 feet.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Whether procurement is for life preservers support packages or life preservers only.
(c) Sample for first article inspection (see 3.1).
(d) When group A and B testing is to be witnessed by Government Inspector (see 4.4.1 and 4.4.2).
(e) When examination of resubmitted lots is to be witnessed by the Government Inspector (see 4.6.2.1).
(f) Level of packaging and packing required (see 5.1).

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.2.7 and 3.2.8	Certificate of compliance	DI-E-2121	
4.1.1	Inspection system program plan	DI-R-4803	

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 4220-N281)

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